

Fionn O'Connor

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Dublin, Ireland | Irish citizen (full EU work rights) | Open to relocation

AI Engineer with a First Class Honours MEng in Computer Engineering (Trinity College Dublin) and ~1.5 years deploying production ML, data pipelines and test automation in a regulated manufacturing environment. Strong foundation in statistical anomaly detection, NLP, and end-to-end data infrastructure – comfortable owning a problem from raw data through to production, and working directly with cross-functional stakeholders along the way.

Skills

Programming: Python, SQL, C#, C/C++, JavaScript/TypeScript

ML & Data: Pandas, Scikit-learn, ONNX, spaCy, Hugging Face, TensorFlow/Keras, OpenCV

Cloud & Infra: Microsoft Azure, Databricks, AWS (Lambda, S3, IoT Core), Elasticsearch, Linux, Git

Industrial & Web: Ignition (SCADA), MQTT, React **Spoken:** English (native), French (beginner)

Experience

AI Engineer – Siemens Healthineers March 2025 – Present

- Re-architected a serial inference pipeline into a parallelised Pandas workflow, cutting end-to-end runtime (file read, processing, ONNX inference) by $\sim 9.1\times$ for a 200-file batch (38.6s $\rightarrow$ 4.2s).
- Owned two epics spanning kickoff, development, integration and verification to capture and analyse new trace data from field instruments, closing a 20-year data gap; selected as subject matter expert.
- Integrating automated tests into Ignition (SCADA) work instructions, scripting one-click test triggers and tester prompts in Python to give engineering live status on a mandatory 2–3 day test stage that logs none today.
- Built statistical anomaly and drift detection on instrument trace data, run daily by a team across manufacturing and field instruments; has caught both sudden spikes and slow field drifts, aiding diagnosis.
- Developing a spaCy NER model to detect PII in field service notes for EU compliance (~ 400 hand-labelled samples).

Research Assistant – Trinity College Dublin

Aug 2024 – March 2025

- Ran EEG experiments with 20+ participants for a PhD research project, then processed the recorded signals in MATLAB, marking trigger onsets for analysis.

Earlier Experience

Software Engineering Intern – Mawla Agency, Remote

Aug 2022 – Oct 2022

Built React/Next.js and TypeScript components for client web apps; contributed to design reviews and client docs.

Technology Consulting Intern – EY, Dublin

Jun 2022 – Jul 2022

Supported the PMO on a multi-team engagement; built a Confluence template for the Solution Architecture team.

Education

MEng Computer Engineering, First Class Honours (1.1) – Trinity College Dublin Sept 2019 – May 2024

- Relevant coursework: deep learning, probability and statistics, data structures and algorithms, information retrieval

Erasmus Exchange, MSc Artificial Intelligence programme – KU Leuven, Belgium

Feb 2023 – Jul 2023

- Completed MSc AI modules: artificial neural networks, speech recognition, computer vision, SVMs

Notable Projects

PartyBot – IoT Speech-Recognition Party Game, Team of 6

- Built the AWS cloud integration end-to-end: 3 event-driven Lambda functions and 2 S3 buckets pushing UI-selected word lists to on-device model updates over IoT Core/MQTT, plus live per-player detection stats.
- Tuned on-device speech recognition (WakeNet, MultiNet), improving accuracy and fixing a default-language defect.
- Tools Used: AWS (IoT Core, Lambda, S3), MQTT, ESP32, React

FurniShare – Second-hand Furniture Marketplace App for iOS and Android

- Led team of 22 engineers developing a marketplace app over the course of 8 weeks.
- Tools Used: Firebase, React Native, Figma, Git, Expo Go

Masters Research Project – EEG study of humour perception using temporal response functions (MATLAB, Praat).

Hobbies and Interests

- Interests: Climbing, music, fiddle player, céilí band, football, scuba diving